

HSE TOOLBOX TALK

EXCAVATION AND TRENCH SAFETY

PURPOSE AND SCOPE

Prevent trench collapse, utility strikes, falls and equipment hazards during excavation work. Verify permits and underground utilities before excavation.

KEY HAZARDS

1



CAVE-IN

2



UTILITY STRIKE

3



FALL

4



WATER

5



ATMOSPHERE

6



EQUIPMENT

7



FALLING MATERIAL

8



ACCESS

CRITICAL VALUES TO VERIFY IN THE FIELD

1.52 m

General protective-system trigger

7.6 m

Maximum lateral travel to ladder

0.61 m

Minimum spoil setback



REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

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STANDARDS AND FIELD REQUIREMENTS

OSHA 1926 Subpart P requires competent-person inspection, utility location, access, protective systems, water and atmosphere controls.

OSHA 1926.651(b)

UNDERGROUND UTILITIES

Locate and mark all services before excavation and verify them by a safe method when approaching.

OSHA 1926.651(c)

SAFE ACCESS

In trenches 1.22 m or deeper, keep workers within 7.6 m lateral travel of a ladder.

OSHA 1926.652(a)

PROTECTIVE SYSTEM

At 1.52 m or deeper, provide sloping, shoring or shielding unless the excavation is entirely stable rock.

OSHA 1926.651(k)

COMPETENT-PERSON INSPECTION

Inspect daily before work and again after rain or any condition change.

MINIMUM EVIDENCE BEFORE START

- | | |
|--|--|
| ☐ Excavation permit and layout | ☐ Utility-detection record |
| ☐ Soil classification | ☐ Shoring / sloping / shield design |
| ☐ Daily competent-person inspection | ☐ Emergency and dewatering plan |

SUPERVISOR DISCUSSION POINTS

- Is the permit approved?
- Are utilities identified?
- Is trench protection installed?

TRENCH PROTECTIVE-SYSTEM INSPECTION



Shield / shoring - Ladder - Spoil setback - Barrier - Utility marking

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. AVOID OR REDUCE EXCAVATION

Verify permit. Locate underground services.

2. VERIFY SERVICES AND SOIL

Use trench protection. Provide ladders. Barricade excavation.

3. INSTALL PROTECTIVE SYSTEM

Keep spoil away from edge. Maintain equipment clearance. Inspect after rain.

4. CONTROL ACCESS AND CONDITIONS

Gas test if required. Prepare emergency plan.

CRITICAL WARNING



Never enter an unprotected excavation.

SITE CHECKLIST (BEFORE STARTING WORK)

- ☐ Permit approved

☐ Protection installed

☐ Barricaded

☐ Gas checked

☐ Standby ready
- ☐ Utilities identified

☐ Ladder available

☐ Ground stable

☐ Equipment clear

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EXCAVATION AND TRENCH SAFETY

ATTENDANCE AND APPROVAL

Project / Site:

Date:

Presented by:

Work area:

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			